

Using Conceptual Models in Interaction Design

- Interaction design consists of concepts, task flow, and presentation.
 - Concepts
 - e.g., all concepts that the application's user interface exposes to users
- -Task flow
 - e.g., the sequence of operations that users execute to accomplish tasks
- - Presentation
 - e.g., the controls, displays, etc. that comprise its user interface

Using Conceptual Models in Interaction Design (Contd.)

- A conceptual model is a high-level description of an application.
- It enumerates all concepts in the application that users can encounter, describes how those concepts relate to each other, and how those concept it into tasks that users perform with the application.
- More precisely, a conceptual model of an interactive application includes, the data “objects,” which a user may encounter, and the “operations, attributes, and relationships, which a user may perform on the data objects.
- It’s important to note that the conceptual model is independent of task fl ow and presentation.
- As you work out task flow, you may find problems in the conceptual model, and you may need to modify it.

Using Conceptual Models in Interaction Design (Contd.)

- Lists are not models.
- Sentences are not efficient means of describing structure - Text is often redundant.
- **Node-link diagrams—concept maps—are a more efficient means of representing conceptual models**

Example of a conceptual model Johnson + Henderson – Alarm Clock

- The clock **stores** the ***current time*** of day,
- continually **updating** it to track the passage of time.
- It **displays** the current time constantly.
- ***Users*** can **set** the current time.
- ***Users*** can set an **alarm** at a ***specified time***, or no alarm.
- When an alarm is set and the current time equals the set **alarm time**, the **alarm** is **triggered**.
- Users can **turn off** an alarm.

Example of a conceptual model Johnson + Henderson – Alarm Clock (Contd.)

- could be expressed in digital or analog travel clocks.



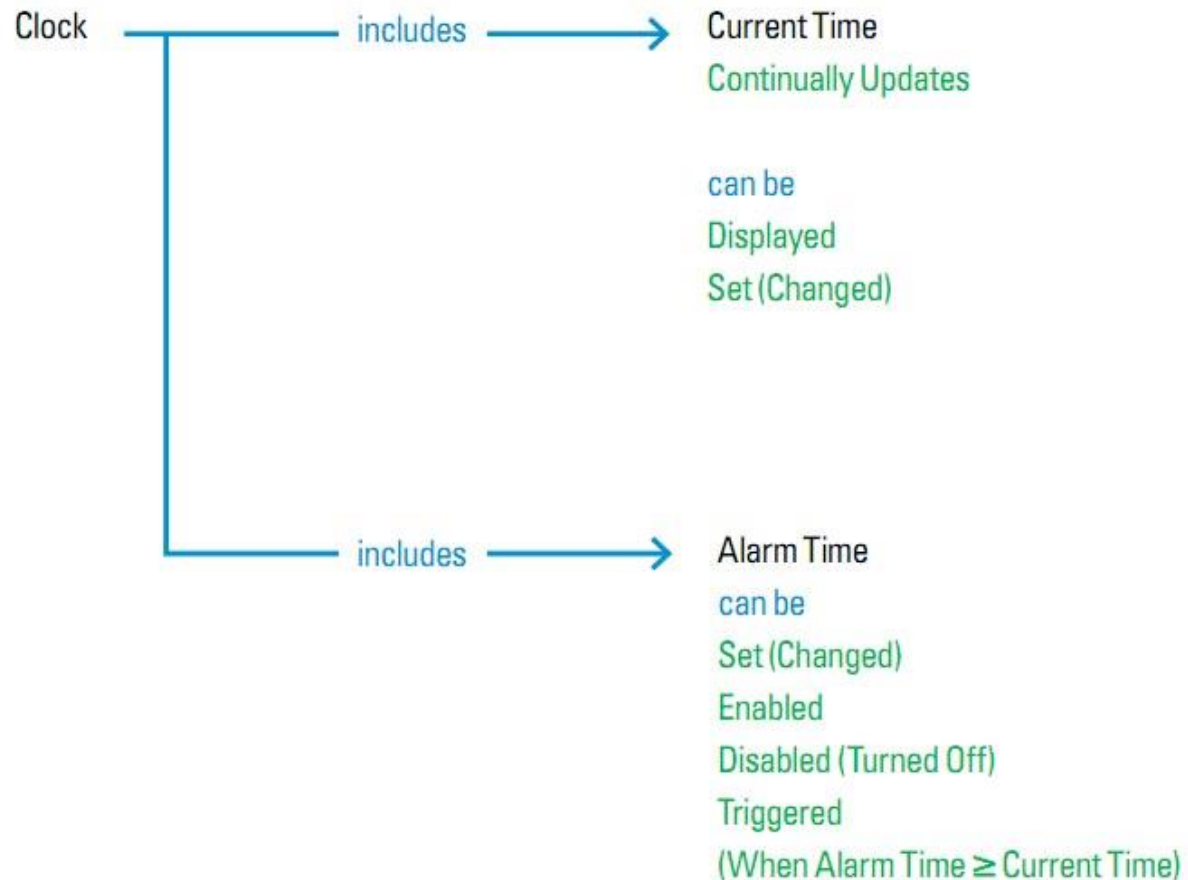
Example of a conceptual model Johnson + Henderson – Alarm Clock (Contd.)

- could also describe the iPhone and Galaxy Nexus clocks.



Example of a conceptual model Johnson + Henderson – Alarm Clock (Contd.)

- A conceptual model of an alarm clock represented as a concept map.



Example of a conceptual model Johnson + Henderson – Alarm Clock (Contd.)

- Conceptual model of an alarm clock with detail added.



Summary

- Product teams should agree on a conceptual model—a definition of what users need to know—well before creating wire-frames and writing code.
- The model will change over the course of the product development process.
- Developing a standard form for conceptual models will make them easier to understand, easier to make, and easier to teach new designers.

References & Further Reading

- http://www.dubberly.com/presentations/conceptual_models.pdf
- Conceptual Models: Begin by Designing What to Design by Jeff Johnson and Austin Henderson